Question ID: 68f343c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

Answer

A. -4

B. 8

C. 42

D. 44

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(x) = x + 6$. Substituting 2 for x in the function f yields $f(2) = 2 + 6$, or $f(2) = 8$. It's also given that $g(x) = 6x$. Substituting 2 for x in the function g yields $g(2) = (6)(2)$, or $g(2) = 12$. Substituting 8 for $f(2)$ and 12 for $g(2)$ in the expression $7f(2) - g(2)$ yields $7(8) - 12$, or 44. Therefore, the value of $7f(2) - g(2)$ is 44.

Choice A is incorrect. This is the value of $f(2) - g(2)$, not $7f(2) - g(2)$.

Choice B is incorrect. This is the value of $f(2)$, not $7f(2) - g(2)$.

Choice C is incorrect and may result from conceptual or calculation errors.